AN AI BASED DENGUE OUTBREAK PREDICTION AND INTERVENTION OPTIMIZATION SYSTEM FOR THE COLOMBO DISTRICT SRI LANKA

COMMON INTEGRATED SOLUTION REPORT

Group ID R26-IT-061

Team member	Student ID	Individual component
Keenavinna K.A.A.I.U	IT22214034	Machine learning prediction
Hettiarachchi A.T	IT22165848	Explainable artificial intelligence
Wickramarachchi T.T.	IT22243980	Intervention optimization
Adithya A A H	IT22281982	PHI inspection and multilingual health intelligence

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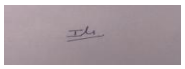
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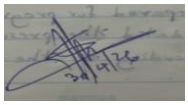
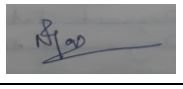
DECLARATION AND SUPERVISOR CERTIFICATION

We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use its content in future work.

Generative AI assistance in preparing this report is described in Appendix E. We remain responsible for its accuracy, originality, citations, technical evidence, contribution account, and final disclosure under the applicable module policy.

Student ID	**Signature**	**Date**
IT22214034		2026.10.11
IT22165848		2026.10.11
IT22243980		2026.10.11
IT22281982		2026.10.11

Supervisor certification. The above candidates carried out this research project under my supervision.

Role and name	**Signature**	**Date**
Supervisor Ms. Shashika Lokuliyana		2026.10.11
Co-supervisor Ms. Narmada Gamage		2026.10.11

ABSTRACT

Dengue decision support requires a connection between risk estimation, model interpretation, intervention planning, field observation, and understandable health information. This group project develops an AI based dengue outbreak prediction and intervention optimization system for the Colombo District, Sri Lanka, through four complementary components. A weekly data-engineering and Random Forest pipeline provides district risk classification; SHAP supplies global and local model explanations; a proposed Q-learning agent connects valid MOH-area risk states to advisory interventions and PHI warnings; and a field-and-language component supports image inspection and evidence-oriented multilingual queries. The common architecture combines Flutter, backend API services, and Supabase-supported authentication and data functions. Evaluation separates model performance, interface consistency, workflow reliability, and operational readiness. The prediction thesis reports 84.31% accuracy and 82.11% weighted F1 on unseen 2025 data after training on 2022–2024. A different supporting prediction experiment in the XAI thesis reports 86.35% accuracy and is retained separately pending model-provenance reconciliation. Recorded NLP experiments show strong controlled context and misinformation classification, moderate retrieval, and weak overall severity classification on a 32-case holdout. A structured Sinhala API response demonstrates a selected backend integration path. The intervention component remains preliminary; detector benchmarking, formal group stakeholder validation, and complete cloud operation are outstanding. The principal contribution is a common, human-reviewed architecture with explicit component contracts and evidence boundaries. The findings support a research prototype with measured subpaths, rather than a claim of complete operational effectiveness or reduced dengue incidence. Future work should consolidate the selected model release, verify area-to-action integration, strengthen field and language evaluation, and execute common acceptance and deployment tests.

Keywords: dengue decision support, outbreak prediction, SHAP, intervention planning, multilingual health intelligence

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CONTENTS

DECLARATION AND SUPERVISOR CERTIFICATION i
ABSTRACT ii
ACKNOWLEDGEMENTS iii
LIST OF FIGURES vi
LIST OF TABLES vii
LIST OF ABBREVIATIONS viii
1 INTRODUCTION AND PROJECT CONTEXT 1
1.1 Background and Context 1
1.1.1 Prediction as a Research and Operational Foundation 2
1.1.2 Interpretation and Action Within a Common Workflow 2
1.2 Stakeholders and User Needs 3
1.3 Alignment With the Sustainable Development Goals 4
1.4 Purpose and Scope of the Common Report 5
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 7
2.1 Overall Problem Definition 7
2.2 Root Causes and Overall Research Gap 7
2.3 Subproblem and Component Mapping 9
2.4 Overall Aim and Group Objectives 10
2.5 Scope Assumptions and Constraints 11
3 OVERALL SOLUTION STRUCTURE 13
3.1 Integrated Solution Overview 13
3.2 High Level Architecture 14
3.3 Component 1 Summary 15
3.4 Component 2 Summary 16
3.5 Component 3 Summary 17
3.6 Component 4 Summary 17
3.7 End to End User and Data Flow 18
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 21
4.1 Integration Strategy 21
4.2 Interfaces Data Contracts and Shared Models 22
4.3 Shared Nonfunctional Requirements 24
4.4 Integrated Workflow and Exception Handling 25
4.5 Team Contribution and Dependency Matrix 27
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 29
5.1 Common Technology Stack and Infrastructure 29
5.2 Development and Integration Process 30
5.3 Deployment and Operational Configuration 31
5.4 Evidence of Team Execution 33

6 INTEGRATED TESTING EVALUATION AND VALIDATION 35
6.1 Evaluation Framework and Success Criteria 35
6.2 Integration and End to End Testing 36
6.3 Overall Results 38
6.3.1 Prediction Results and Model Provenance 38
6.3.2 Explanation Results 39
6.3.3 Multilingual Classification and Retrieval Results 39
6.3.4 Severity and Field-Support Limitations 41
6.3.5 Integrated Maturity and Deployment Findings 41
6.4 Stakeholder and User Validation 42
6.5 Achievement of Common Objectives 43
6.6 Integrated Limitations and Threats 44
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION . 46
7.1 Value Proposition and Beneficiaries 46
7.2 Market Alternatives and Differentiation 46
7.3 Business Model and Feasibility 48
7.4 Adoption Intellectual Property and Compliance 49
7.5 Sustainable Development Goal Contribution 50
8 CONCLUSION AND FUTURE DIRECTIONS 52
8.1 Overall Conclusion 52
8.2 Principal Integrated Contribution 53
8.3 Future Directions 54
REFERENCES 56
APPENDICES 58
Appendix A Integrated Interface Specification 58
Appendix B Team Contribution Declaration 60
Appendix C Integrated Test Matrix 61
Appendix D Prospective Stakeholder Validation Protocol 63
Appendix E Academic Integrity and AI Assistance 65
Appendix F Common Model and Evidence Reconciliation Record 66

LIST OF FIGURES

Figure 3.1 Common architecture and intervention design dependency 14
Figure 3.2 Target professional workflow and validation gates 19
Figure 4.1 Intended warning transaction states and partial failure 26

LIST OF TABLES

Table 1.1 Stakeholder tasks and validation needs 4
Table 2.1 Four subproblems and component ownership 9
Table 2.2 Group objectives and evaluation evidence 10
Table 4.1 Component integration dependencies 22
Table 4.2 Documented route names and evidence status 22
Table 4.3 Research responsibilities and shared dependencies 27
Table 5.1 Technology layers and implementation evidence 29
Table 5.2 Recorded field and language artefact sizes 30
Table 5.3 Execution evidence and interpretation 33
Table 6.1 Group evaluation questions and acceptance evidence 36
Table 6.2 Prediction experiment records retained separately 38
Table 6.3 Recorded NLP task results 39
Table 6.4 Integrated capability and evidence status 41
Table 6.5 Objective achievement against current evidence 43
Table 6.6 Main validity threats and required evidence 45
Table 7.1 Functional alternatives and proposed differentiation 47
Table A.1 Logical interface records and validation rules 58
Table B.1 Component allocation for member confirmation 60
Table B.2 Member confirmation signatures 60
Table C.1 Prospective group acceptance and exception tests 61
Table F.1 Outstanding common evidence reconciliation 66

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
C1–C4	Research components 1 to 4
F1	Harmonic mean of precision and recall
HTTP	Hypertext transfer protocol
mAP	Mean average precision
MOH	Medical Officer of Health
MRR	Mean reciprocal rank
NDCU	National Dengue Control Unit
NLI	Natural-language inference
NLP	Natural-language processing
PHI	Public Health Inspector
QA	Question answering
RAG	Retrieval-augmented generation
RF	Random Forest
RL	Reinforcement learning
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
UI	User interface
WHO	World Health Organization
XAI	Explainable artificial intelligence

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Dengue prevention requires a continuous connection between surveillance, interpretation, public-health action, field observation, and understandable information. In the Colombo District, the research problem is therefore broader than estimating a risk category. A useful decision-support system must help a responsible user understand the estimate, identify an appropriate next step, and review the evidence needed for that step. The integrated project brings these activities into a common mobile application while preserving the different responsibilities of statistical prediction, model explanation, intervention planning, and field support.

The National Dengue Control Unit describes the Western Province, including Colombo, as an important dengue setting and identifies seasonal transmission, urban development, population density, and environmental conditions as relevant contextual factors [1]. These conditions motivate a district-focused research prototype. They do not establish that every location within Colombo has the same risk or that a district-level estimate can be directly applied to an individual premise. This distinction shapes both the system architecture and the evaluation strategy.

The public-health workflow crosses several information scales. Historical dengue and meteorological records describe patterns over time. A Medical Officer of Health (MOH) area provides an operational unit for planning. A Public Health Inspector (PHI) observes specific premises and potential breeding environments. A public information user asks a question in everyday language. The same application can connect these scales, but it must retain their meaning rather than convert all outputs into a single undifferentiated risk score.

The project is titled An AI Based Dengue Outbreak Prediction and Intervention Optimization System for the Colombo District Sri Lanka. Its four research components are machine learning prediction, explainable artificial intelligence, intervention optimization, and PHI field inspection with multilingual dengue health intelligence. The integrated solution is designed as an advisory platform. It supports professional interpretation and public information access; it does not establish

clinical diagnosis, confirm vector breeding from an image alone, or autonomously execute interventions.

1.1.1 Prediction as a Research and Operational Foundation

The prediction component prepares a weekly dataset combining dengue history with meteorological and temporal variables. Its Random Forest model classifies the target outbreak level as Low, Medium, or High. The component thesis reports training on 2022–2024 and testing on the subsequent 2025 period, with 84.31% accuracy and 82.11% weighted F1 [2]. Chronological testing is relevant because the intended application uses past observations to support decisions about later periods. It is more informative for that purpose than a result obtained only by mixing nearby observations across a random split.

Random Forest combines multiple decision trees and is a suitable methodological foundation for structured classification [3]. The project's choice is also compatible with a Python backend and tree-based explanation tools. Suitability does not imply that it outperforms every alternative model. No common benchmark comparing all candidate algorithms under identical data and temporal conditions is established here. The reported result is interpreted within the feature configuration, target definition, and test period of the prediction component.

Two prediction contexts must remain distinct. Historical or observed-period requests can use recorded inputs. Requests beyond the observed period use historical seasonal information where direct future weather observations are unavailable [2]. Those future inputs are assumptions about the requested period. A risk estimate based on them is a model-based scenario, and the observed-period test accuracy is not a measured accuracy for every future scenario.

1.1.2 Interpretation and Action Within a Common Workflow

Explainable artificial intelligence (XAI) addresses the gap between receiving a label and understanding the model's behaviour. SHapley Additive exPlanations (SHAP) assigns feature contributions to an individual model output [4]. The XAI component uses this principle for global and local explanations [5]. Global analysis summarises

contributions across observations, while local analysis explains a selected prediction. A contribution identifies a relationship within the model; it does not show that changing the feature would cause a corresponding change in real dengue transmission.

The intervention component addresses a different question: what advisory action should follow a defined risk condition? Its proposed Q-learning formulation represents risk context as a state, intervention options as actions, and feedback as a reward [6]. Q-learning provides a framework for updating state-action values through experience [7]. However, the project's state encoding, action catalogue, reward function, and measured policy behaviour remain to be verified. Consequently, the intervention pathway is described as a design contribution rather than as demonstrated epidemiological optimization.

Field inspection and health information complete the operational context. An image detector draws attention to configured environmental objects for an officer to inspect, while a multilingual pipeline routes questions, retrieves evidence, and applies consistency checks [8]. Neither output replaces the relevant human judgement. An environmental object can be dry, covered, or otherwise managed; a fluent answer can still depend on incomplete evidence. The combined platform therefore needs explicit uncertainty and review at each consequential boundary.

1.2 Stakeholders and User Needs

The principal professional users are PHIs and MOH teams who need timely, interpretable information for reviewing local prevention activities. Programme managers are secondary users who may need an overview of risk, system status, and warning history. Public information users need dengue-related material in language they can understand. The student development team and academic supervisors are research stakeholders responsible for assessing the artefact, its evidence, and the limits of its claims.

These needs are derived from the component problem analyses and the documented public-health setting [1], [2], [5], [6], [8]. A completed formal group stakeholder study is not established. Accordingly, the needs below are design requirements, not a

claim that a representative sample endorsed the product. Table 1.1 distinguishes each stakeholder's task and the evidence needed to validate the proposed support.

Table 1.1 Stakeholder tasks and validation needs

Stakeholder	Task supported	Validation needed
PHI	Review image detections and area warnings	Field task observation and false-negative review
MOH team	Interpret risk and review proposed interventions	Area provenance and recommendation appropriateness
Programme manager	Review service outputs and warning records	Audit completeness and operational reliability
Public information user	Ask questions in supported language forms	Comprehension and evidence-grounding assessment
Research team	Reproduce experiments and maintain interfaces	Versioned data models and repeatable tests

Professional and public access have different purposes. A public information user should not gain officer privileges merely by registering an account. Conversely, an authenticated officer needs more than an accessible screen: the backend must enforce authorization for restricted functions. The field-support thesis describes an authorized-employee concept and an authentication gate within the shared application [8]. These provide an integration pattern, while a complete security test remains necessary to demonstrate protection at every service boundary.

Usability also depends on task context. A PHI may be outdoors, using a small screen, and dealing with variable connectivity and image quality. A public user may write in Sinhala, English, or Singlish, with spelling variation and incomplete context. The current NLP evidence explicitly includes these three language forms [8]. Tamil is an important future requirement, but a completed Tamil benchmark is not established and must not be implied by the term multilingual.

1.3 Alignment With the Sustainable Development Goals

The principal alignment is with Sustainable Development Goal 3. Target 3.3 concerns communicable diseases and neglected tropical diseases; Target 3.d concerns capacity for early warning, risk reduction, and management of health risks

[9]. The project contributes at the level of a research tool intended to support those activities. Its prediction, interpretation, and field-support functions can provide a basis for more organized information use, but a software prototype does not itself demonstrate progress against population health indicators.

The intended contribution depends on adoption and evidence. An accurate model that is unavailable during field work has limited practical value. An accessible answer that lacks relevant evidence may be misleading. An intervention recommendation that ignores local resources may not be usable. For this reason, the SDG relationship is evaluated through intermediate outcomes such as task completion, understanding, traceability, and service reliability before any claim about disease outcomes.

Local-language access is also a responsible design objective. It can reduce an information barrier for some users, provided that terminology and meaning are assessed by competent speakers and domain reviewers. The project does not equate interface translation with equal performance. A future study should compare comprehension and response quality across language forms and should report the availability of functions under realistic device and connectivity conditions.

1.4 Purpose and Scope of the Common Report

This report presents the group problem, the four-component architecture, component dependencies, shared implementation context, integration requirements, and the combined evaluation. The individual theses remain the detailed accounts of each research contribution [2], [5], [6], [8]. The common report focuses on what happens between components and what can reasonably be concluded about the complete system.

The evidence supports a prototype with several reported implementations and measured component experiments. It does not establish a fully validated four-component public-health service. In particular, the intervention component remains preliminary, the vision benchmark is unreported, formal group stakeholder validation is outstanding, and full multi-model cloud operation remains resource-constrained. These are substantive research limitations rather than reasons to combine unrelated results into a single system accuracy.

The report also retains differences between prediction experiment descriptions. The prediction thesis reports a 2022–2024 training period and 84.31% accuracy; the XAI thesis cites a 2015–2024 model experiment and 86.35% accuracy, together with an area-week application [2], [5]. Chapter 6 separates those records. Neither value is averaged or silently substituted for the other. A common release must identify the exact model, feature schema, geographical unit, and dataset before a single integrated benchmark can be declared.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

The common problem is the fragmentation of dengue decision support across risk estimation, interpretation, intervention planning, field observation, and public information. A risk label alone does not explain the model's reasoning, identify a locally appropriate response, verify a premise, or answer a user's question. When these tasks are disconnected, a professional must transfer context manually between tools and determine whether apparently related outputs concern the same place, time, data source, and model version.

The problem is especially important where the analytical and operational units differ. The current prediction implementation is described as a Colombo District model, whereas the intervention design consumes MOH-area risk context [2], [6]. A separate area proxy dataset cannot automatically make the district classifier an independently validated MOH-area model. Without clear provenance, an area label may imply greater geographical precision than the research evidence supports.

The overall research question is therefore whether a common mobile and backend architecture can connect these four forms of assistance while preserving data meaning, model consistency, human review, and traceable evidence. This formulation makes integration quality part of the research problem. It also allows a qualified conclusion: component-level capabilities can be demonstrated even when an operational linkage or group-level validation remains incomplete.

2.2 Root Causes and Overall Research Gap

The first root cause is variation in data granularity and availability. District observations, MOH-area proxy estimates, field images, and language corpora represent different constructs. Their periods and completeness also differ. The prediction thesis describes weekly district modelling data for 2022–2025 and a separate MOH proxy/context dataset for 2015–2026 [2]. These sources need separate

identity and provenance throughout the application rather than being treated as interchangeable inputs.

The second cause is the separation between algorithm evaluation and workflow evaluation. Classification accuracy measures agreement with labels in a defined dataset. It does not measure whether the mobile interface shows the correct week, whether an explanation belongs to the displayed prediction, whether an officer can recover from a network error, or whether a warning reaches the intended recipient. An integrated research gap remains when those boundaries are described but not tested with repeatable records.

The third cause is limited interpretation. A Random Forest prediction can be useful without exposing its feature contributions. SHAP provides a way to explain the model output, but the explanation introduces further requirements: correct feature order, class identity, output space, and model version. A mismatch can create an apparently convincing explanation of a different prediction. Interpretation therefore needs both analytical correctness and interface consistency [5].

The fourth cause is the distance between risk and action. Public-health planning uses local knowledge, available resources, evidence, and professional responsibility. A Q-learning recommendation cannot acquire those qualities merely by using an action-value table. The intervention research needs an operationally meaningful state and action space and a reward whose meaning is documented. Successful email submission is a communication outcome and must not be treated automatically as a reward for an effective dengue intervention [6].

The fifth cause is uneven information quality across languages and tasks. The field-support component combines context classification, severity signals, retrieval, natural-language inference, misinformation classification, and numeric checking [8]. Their recorded performance is uneven. A strong context score cannot compensate for weak urgency classification or irrelevant retrieval. Multilingual transfer methods provide useful representations [10], but local performance and wording still need task-specific evaluation.

The overarching gap is thus a traceable connection from data to prediction, prediction to explanation, risk context to reviewed action, and field or information

requests to evidence. The project addresses that gap through shared interfaces and bounded component responsibilities. Its contribution is strongest where the connections and outputs are documented; claims of field effectiveness require a later operational study.

2.3 Subproblem and Component Mapping

The four subproblems are complementary. The prediction component estimates risk from structured records. XAI explains the model's estimate. Intervention optimization proposes an advisory action for a valid operational state. PHI and multilingual support assist observation and information access. Table 2.1 identifies these responsibilities without assigning a second research owner to the same core method.

Table 2.1 Four subproblems and component ownership

Component and owner	Subproblem	Input and principal output
C1 Keenavinna K.A.A.I.U	Historical and environmental data need a future-period risk classification	Weekly features to Low Medium or High risk
C2 Hettiarachchi A.T	Users cannot interpret a prediction from its label alone	Same model and input to SHAP contributions
C3 Wickramarachchi T.T.	A risk condition does not specify a reviewed response	Valid MOH state to proposed intervention and warning
C4 Adithya A A H	Field evidence and local-language information lack a unified support path	Image or text to detections or grounded information

C1 and C2 have the closest analytical dependency. An explanation must use the model and feature vector associated with the exact prediction being displayed. C3 depends on a different contract: it needs a validated operational risk state with area provenance. C4 shares the mobile application, backend conventions, and officer authorization context; it can answer a public information query without waiting for a prediction or recommendation.

All four components are necessary for the proposed complete solution, but they need not execute sequentially for every user. A user may request a district prediction and its explanation, an officer may submit a field image, or a public user may ask a

prevention question. The common platform connects these paths through shared identity, navigation, and evidence conventions. It must not introduce an unsupported dependency, such as automatically retraining the prediction model from unverified image detections.

2.4 Overall Aim and Group Objectives

The overall aim is to develop and evaluate an integrated, human-reviewed dengue decision-support prototype for the Colombo District that connects risk classification, model explanation, intervention planning, PHI field assistance, and multilingual information through a common application and backend architecture.

Group objectives are defined against observable artefacts and outcomes. Table 2.2 links each objective to its component responsibilities and verification method. These objectives avoid presuming a reduction in dengue cases, which would require prospective outcome data and a suitable comparative design.

Table 2.2 Group objectives and evaluation evidence

ID	Objective	Evidence and component link
G1	Prepare a temporally evaluated risk pipeline	C1 dataset split and held-out classification results
G2	Associate explanations with the correct prediction	C1 C2 feature mapping model identity and explanation checks
G3	Define reviewed risk-to-action support	C3 state action reward specification and warning tests
G4	Provide field and multilingual assistance	C4 detector path NLP benchmarks and API response
G5	Integrate functions through a common application	Navigation authorization schemas and end-to-end scenarios
G6	Establish readiness and responsible limits	Resource tests stakeholder study and evidence provenance

G1 is assessed using the specified temporal holdout rather than an arbitrary success threshold selected after seeing the result. G2 requires a repeatable consistency check and an explanation record associated with a prediction identifier. G3 requires both

algorithm evidence and communication evidence; one cannot replace the other. G4 combines model-level and workflow evidence while retaining separate scores for each task.

G5 and G6 are the principal common-report objectives. A shared application shell provides part of the evidence for G5, but full achievement requires verification across the four components, including exception paths. G6 requires a documented account of what can run, what has been measured, and what remains dependent on future work. Chapter 6 maps the available findings to these objectives and identifies partial achievement where appropriate.

2.5 Scope Assumptions and Constraints

The geographical focus is Colombo District. The principal predictive task is three-class weekly risk classification, rather than individual diagnosis or a numerical case-count forecast. SHAP explanations describe model behaviour. PHI detections describe configured visual objects, rather than confirmed larvae or statutory compliance. The multilingual service provides dengue-specific information, with English, Sinhala, and Singlish evidenced in the component evaluation. Tamil remains a future validation requirement [2], [5], [8].

The system assumes access to historical modelling records, compatible trained artefacts, a mobile device, and network connectivity when server inference is needed. Future-period requests additionally depend on seasonal assumptions where observed weather inputs are unavailable. The quality and timeliness of upstream information constrain every downstream function. Missing observations should produce a clear unavailable or incomplete state, not an unexplained default risk label.

The intervention design assumes an MOH-area risk condition, a finite intervention catalogue, and feedback suitable for Q-value updates [6]. These assumptions are not yet equivalent to an implemented and evaluated learning environment. In particular, the district-to-area relationship needs confirmation. An area proxy remains contextual unless a verified mapping and evaluation establish its role in action selection.

Infrastructure is another constraint. The field-support thesis records NLP model artefacts totalling approximately 3.7 GB, while its vision checkpoint is about 5.22 MB [8]. File size is not a full measurement of runtime memory, but it makes a 512 MB hosting assumption implausible for loading the complete stack. The common design must distinguish a reported prediction-service deployment from a fully operational deployment of every model and database dependency.

The report excludes claims of clinical efficacy, automatic enforcement, optimal resource allocation in real practice, reduced incidence, and formal compliance certification. None is established by the current prototype evidence. It includes design requirements for reliable and responsible operation because those requirements determine how the research can be extended and evaluated.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

The integrated solution combines a Flutter application, Python-based inference services, a FastAPI communication layer, and Supabase-supported authentication and data functions. The application is the common point of access. The backend connects requests to bounded services and returns structured outputs for presentation. Each component retains its research method and versioned artefacts, while shared conventions preserve context across the workflow [2], [5], [6], [8].

The central value proposition is continuity of information. A user should be able to review a prediction and its explanation without losing the requested period or model identity. A professional considering a recommendation should be able to inspect the relevant area and risk provenance before accepting it. An officer submitting an image should receive interpretable detections and retain responsibility for physical inspection. A person asking a question should receive an answer associated with retrieved evidence and an appropriate uncertainty state.

This architecture is modular because the four research methods have different input types, resource demands, and maturity. Prediction and SHAP operate on tabular model context. Q-learning requires a discrete decision formulation. Vision requires image decoding and object detection. NLP requires text normalization, classification, embeddings, retrieval, and evidence checks. Modularity supports independent testing and permits a limited service to remain available when a different service cannot execute.

The common system is therefore an integrated prototype architecture with unevenly validated pathways. Prediction-to-mobile and selected NLP backend paths have reported evidence. SHAP integration is described in the XAI thesis. The intervention pathway is a prospective design. Figure 3.1 explicitly marks that boundary rather than representing every box as a proven deployed service.

3.2 High Level Architecture

Figure 3.1 shows the common application and service relationships. Solid connections indicate reported component integration patterns. The intervention link is shown as a design dependency because its full implementation and test record remain outstanding. The diagram is an architecture synthesis; it does not certify a single deployed environment or imply that all services ran simultaneously.

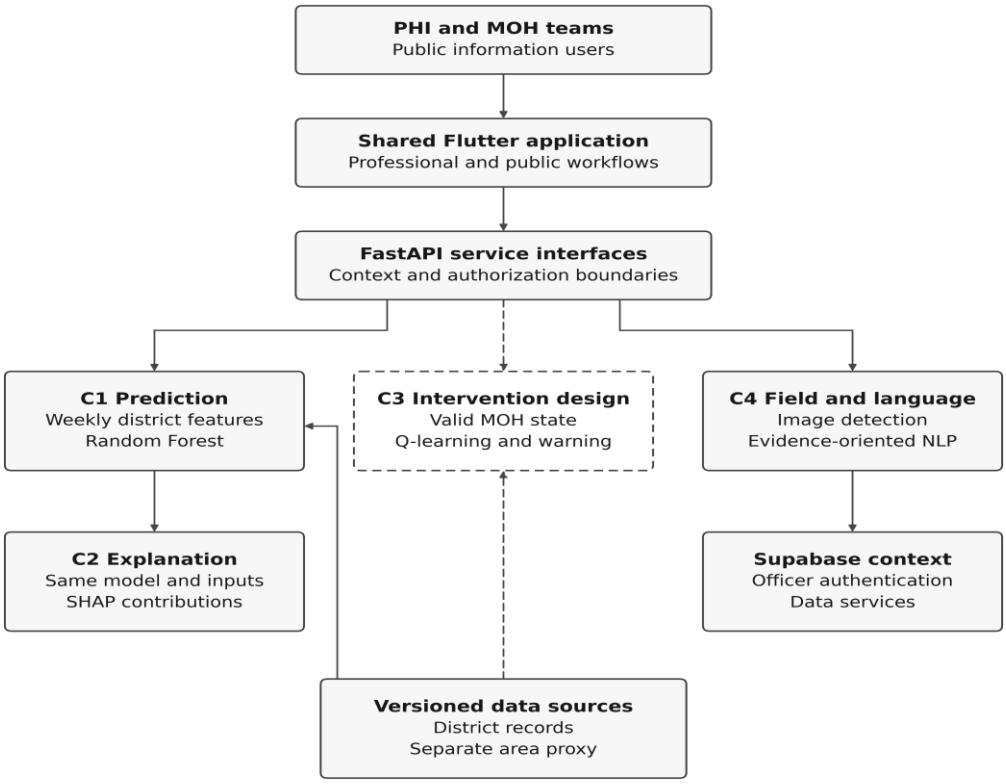


Figure 3.1 Common architecture and intervention design dependency

The application layer separates public information and professional workflows. Prediction and XAI services use a common prediction context. The intervention service must validate area risk and recipient mapping before making a warning request. PHI inspection requires an authorized officer path, while health intelligence processes language and evidence metadata. Authentication and storage are shared concerns, but the actual scope of each database record must match its purpose and access rules.

A district modelling dataset feeds the prediction layer. The area proxy/context dataset is a separate source. The image detector and NLP artefacts also have separate provenance. No architecture arrow converts one source into another by assumption. If the system later adds official area-level records or inspection-derived observations, that addition requires a documented data transformation and a new evaluation.

Deployment boundaries need similar care. The prediction thesis describes FastAPI hosting through Hugging Face Spaces [2]. The field-support thesis describes local device integration, Docker packaging, and resource constraints on full cloud operation [8]. These are compatible as separate development and deployment contexts, but they do not demonstrate one complete production release. A future deployment may separate tabular inference from memory-intensive NLP while retaining one logical application.

3.3 Component 1 Summary

C1 is owned by Keenavinna K.A.A.I.U, IT22214034. Its research contribution is data engineering and machine learning-based dengue outbreak risk prediction. Historical dengue and weather records are aligned into weekly observations, temporal and lag features are constructed, and a Random Forest classifier produces Low, Medium, or High risk. The prediction thesis reports a 2022–2024 training period and unseen 2025 evaluation [2].

The reported feature groups include year, week, month, rainfall, humidity, maximum temperature, minimum temperature, mean temperature, dengue lag 1, dengue lag 2, rainfall lag 1, and humidity lag 1. The shorter feature inventory in one source table omits humidity lag 1, while the later implementation list includes it. The common feature contract must therefore be taken from the selected model artefact rather than reconstructed solely from a narrative inventory.

The service supports manual, automated, future, and dashboard-oriented prediction requests. The named routes are /predict, /predict-auto, /predict-future, and /predict-dashboard [2]. These are documented route names, not an independently verified current API specification. The component supplies the predictive context required by

C2 and the district estimate that can be reviewed alongside area context. It does not establish independent MOH-area predictive accuracy.

The principal result is 84.31% accuracy and 82.11% weighted F1 on the reported 2025 holdout. The implementation contribution includes a FastAPI service and Flutter access. Exact test count, final hyperparameters, per-class confusion entries, target thresholds, and model hashes are not established in the common evidence record. Those omissions constrain reproduction and safety-oriented interpretation, especially the assessment of missed High-risk periods.

3.4 Component 2 Summary

C2 is owned by Hettiarachchi A.T, IT22165848. It adds SHAP-based interpretation to the prediction pipeline. The input consists of the trained prediction model, the associated feature vector, and the prediction context. Outputs include feature contributions for a selected prediction and global summaries across observations [5]. The component does not train a second predictor as its main contribution.

The most important dependency is analytical identity. The displayed explanation must refer to the same model version, ordered feature schema, class output, and observation as the displayed risk. A contribution for a High-class output should not be labelled as a contribution to Medium risk. An explanation derived from seasonal estimates should also retain that origin. The application must present both the prediction and the assumptions behind its inputs.

The XAI thesis describes backend formatting and Flutter presentation of global and local information. It also reports supporting prediction metrics from a different experiment: 86.35% accuracy for a 2015–2024 training description, with a separate application to 220 area-week observations in 2026 Weeks 1–22 [5]. Those are prediction-context results. They are not a measured score for explanation faithfulness, user comprehension, or SHAP quality.

The integrated contribution is transparency of model behaviour within the user workflow. It remains necessary to verify the selected model, test numerical attribution consistency, and assess whether intended users interpret the explanation

correctly. A technically available attribution chart can still be misunderstood as a causal account or as certainty about future dengue conditions.

3.5 Component 3 Summary

C3 is owned by Wickramarachchi T.T., IT22243980. The Dengue Intervention Optimization Agent proposes an action-oriented decision-support layer after risk identification. The intended input is MOH-area risk context. The intended method is Q-learning over defined states and intervention actions, with reward feedback updating state-action values. Its intended output is a readable recommendation and, for a High-risk condition, a human-confirmed PHI warning workflow [6].

The preliminary service design names /recommend, /send-warning, /warning-history, and /moh-areas, with /predict-area-risk identified as an upstream risk-context route. These route names remain design-level until confirmed against source code and a running interface. No exact request schema, action catalogue, learning rate, discount factor, reward magnitude, training trajectory, or measured improvement is asserted.

The communication design separates recommendation generation, user confirmation, warning submission, delivery evidence, and history storage. These are distinct outcomes. A successful backend request does not prove delivery to a PHI, and a stored history entry does not prove that an intervention was performed. The interface should preserve those distinctions so that a partial failure cannot be shown as full success.

C3 contributes the framework connecting risk interpretation to advisory action and operational communication. Its maturity is preliminary relative to the measured C1 and NLP experiments. The common report therefore includes its dependencies, proposed interface responsibilities, and evaluation requirements while recording full implementation and validation as outstanding work.

3.6 Component 4 Summary

C4 is owned by Adithya A A H, IT22281982. It combines PHI field inspection and multilingual dengue health intelligence. The inspection path accepts an image, executes an Ultralytics-compatible detector, and returns detection labels, confidence

values, and bounding boxes for officer review. It is an attention-support tool for environmental inspection; no detector benchmark or final image dataset inventory is established in the recorded thesis results [8].

The health-intelligence path accepts dengue-related text and applies context routing, a severity signal, semantic retrieval, clinical natural-language inference (NLI), and auxiliary misinformation or numeric checks. Sentence embeddings provide a basis for semantic search [11], while retrieval-augmented methods connect responses to an external evidence store [12]. Clinical NLI provides a task framework for assessing a statement relative to evidence [13]. These methods inform the architecture without guaranteeing answer correctness.

The component evidence includes English, Sinhala, and Singlish datasets, several model-level experiments, and a structured Sinhala response from /v1/analyze. It also describes Android device communication, a shared Flutter shell, FastAPI orchestration, officer authentication, and Docker packaging [8]. These provide concrete software and integration context, although complete test counts and formal stakeholder measurements remain unreported.

The measured NLP profile is mixed. Controlled context classification is strong, retrieval is imperfect, and the severity holdout has weak overall multiclass performance. This difference is central to the system's responsible design. The application must allow clarification, out-of-scope handling, review, and unavailable states rather than force every query into a confident answer or urgency decision.

3.7 End to End User and Data Flow

For the prediction path, a user selects or provides a prediction period. The backend validates the request, prepares the model's required inputs, and returns the risk class with available probability information. C2 then uses the same model context to compute and format the explanation. The application presents the estimate, period, geographical unit, input origin, and explanatory factors together. A failure in SHAP should be shown as explanation unavailability while preserving a valid prediction, rather than silently replacing it with a different model output.

For the intended action path, an authorized user selects an MOH area and reviews valid area-risk context. C3 validates the state, returns a proposed intervention, and enables the warning workflow where the risk condition and authorization permit it. The user reviews the area, recipient, and action before confirmation. The backend then records the actual transmission outcome and the independent storage outcome. Figure 3.2 presents this sequence as a target workflow, with unresolved area provenance and intervention execution identified as validation gates.

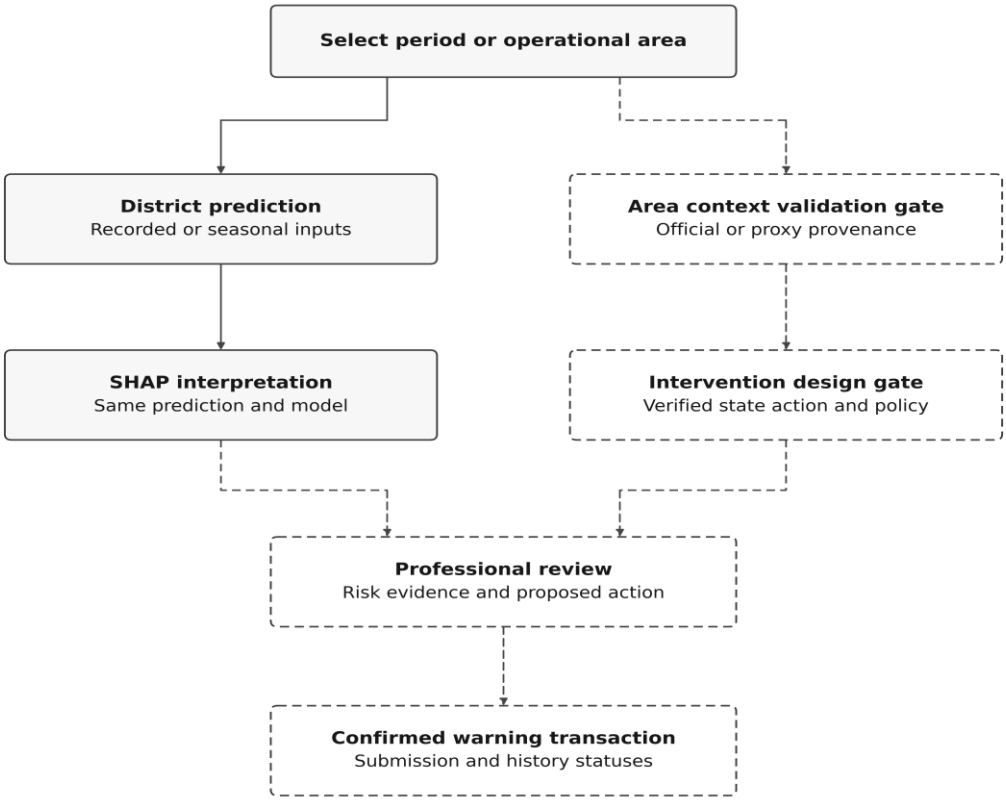


Figure 3.2 Target professional workflow and validation gates

For field inspection, the PHI captures or selects an image and submits it through the authenticated application. The backend validates the file and performs detection. The officer reviews the result against the physical environment. No detection does not imply that the premise is safe. The inspection result is not automatically converted into an official area-risk measurement or a training label.

For health information, a query enters the language-aware pipeline. Insufficient context leads to clarification; out-of-scope content is handled within the application's

domain boundary; a request requiring current data must not be answered from an outdated static corpus as if it were live surveillance. Relevant evidence is retrieved and checked before a structured response is assembled. The Sinhala API example establishes a backend path for this process, rather than complete field or clinical validation.

The shared flow depends on context preservation. Period, area identity, model version, evidence source, and request identity must remain attached to outputs as the user navigates between functions. The system's value comes from that continuity, while each component keeps the limits of its own data and task.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

The integration strategy combines a shared application shell with service boundaries around each research capability. This allows common navigation and authorization without merging every model into one inseparable process. The component theses describe FastAPI-based prediction and health-intelligence services, Flutter integration, and Supabase-supported officer and data functions [2], [5], [8]. The intervention thesis describes the additional recommendation, warning, and history boundaries required for complete operation [6].

Integration is assessed at three levels. Presentation integration establishes that users can access functions through one application. Data integration establishes that outputs retain the correct period, unit, and identifiers. Behavioural integration establishes that a complete task succeeds and that failures are represented accurately. A screenshot of shared navigation is evidence at the first level; it cannot by itself establish the second or third.

The reported field-support development process began with isolated endpoint checks, then connected Flutter screens and tested physical Android communication using a local backend [8]. This is a useful incremental pattern because service defects can be separated from mobile networking and layout defects. It also avoids attributing all failures to model quality. The complete group release should extend this pattern to C1–C2 consistency and the C3 risk-to-warning path.

Interface ownership follows research responsibility. C1 owns prediction input preparation and output semantics. C2 owns the explanation mapping and presentation contract. C3 owns the state/action contract and warning transaction. C4 owns image and text-response adapters. Shared decisions include naming conventions, authentication, application navigation, configuration, and evidence logging. Table 4.1 summarizes the dependencies that require joint review.

Table 4.1 Component integration dependencies

Boundary	Information that must be preserved	Verification responsibility
C1 to C2	Model feature order class and prediction context	Prediction and XAI owners jointly
Risk context to C3	MOH identity period risk and provenance	Prediction and intervention owners jointly
C3 to warning service	Confirmed recipient actions and attempt identity	Intervention owner and shared backend
C4 to application	Image detections or language evidence metadata	Field and NLP owner with application integration
All services	Authorization versions error states and request identity	Group release review

The key risk is premature equivalence: assuming that outputs sharing the word risk concern the same scale and evidence. Integration must reject or clearly label a district-to-area mismatch. A design gate is preferable to silently deriving an intervention state from information whose operational meaning has not been validated.

4.2 Interfaces Data Contracts and Shared Models

The source documents provide named endpoints and conceptual fields, but not a complete common machine-readable API schema. Table 4.2 therefore distinguishes documented routes from design-level routes. Exact HTTP methods, authentication scopes, field names, and deployed versions must be confirmed before these descriptions become a release specification.

Table 4.2 Documented route names and evidence status

Route	Responsibility	Status of evidence
/predict	Manual prediction	Named in C1 thesis
/predict-auto	Automated input preparation	Named in C1 thesis
/predict-future	Future period scenario	Named in C1 thesis
/predict-dashboard	Dashboard prediction context	Named in C1 and C2 documentation
/v1/analyze	Structured dengue text analysis	Recorded Sinhala HTTP 200 example in C4
/predict-area-risk	MOH risk context	Identified in C3 design not confirmed as common contract

/recommend	Advisory action selection	C3 design route
/send-warning	Confirmed warning submission	C3 design route
/warning-history	Warning record retrieval	C3 design route
/moh-areas	Area selection and mapping	C3 design route

The proposed common prediction record contains an immutable prediction identifier, requested year and week, geographical unit, data source type, input values, feature schema version, model version, predicted class, and available class probabilities. The proposed explanation record references that prediction identifier and carries the explained class, output space, feature contributions, and explainer version. These are logical contract requirements; they are not asserted database column names.

An area context record must distinguish observed official information from proxy or estimated information. It should retain MOH identity, time period, source version, completeness, and any mapping to district context. If that mapping is absent or incompatible, the intervention path should remain unavailable rather than borrow a district risk label. This requirement addresses a research limitation shared by C1 and C3 rather than a cosmetic interface issue.

Image and text contracts also need explicit metadata. Detections should preserve the detector version, confidence convention, and coordinate convention. Text responses should retain language, route, context status, evidence identifiers, and available source links. Appendix A expands these logical contracts and their validation rules. The appendix separates recorded fields from proposed common fields to prevent a design document from being mistaken for a verified API dump.

Error handling should distinguish invalid input, unauthorized access, unavailable model, unsupported context, dependency timeout, and partial transaction success. Clients need stable categories rather than arbitrary exception text. A valid response with no detections is different from failed inference; an empty evidence set is different from a successful evidence-grounded answer. The shared model must preserve those distinctions.

4.3 Shared Nonfunctional Requirements

Performance must be evaluated per task. Tabular inference, SHAP calculation, image upload, transformer routing, retrieval, and warning submission have different costs. A single average response time can hide a slow critical path. The group evaluation should measure median and tail latency, startup behaviour, request failure rate, and device task completion under a fixed workload. No completed common latency benchmark is established in the current evidence.

Reliability requires partial availability. If the health-intelligence service cannot load, the application should not imply that prediction is also unavailable when its service is healthy. If SHAP fails, the user should receive a truthful explanation status. If a warning is accepted by the email service but history storage fails, the transaction must retain a partial-success state. Service health and model readiness should be observable separately.

Security requirements cover backend authorization, protected configuration, least privilege, validated uploads, and controlled access to officer records. Hiding a screen is insufficient evidence of authorization. Upload handling must validate that the content is actually decodable and within accepted size and type constraints. Logs should retain enough diagnostic information to reproduce failures without unnecessarily exposing user queries, field images, or contact details.

Privacy is relevant even though the prediction model does not require patient identity. Field photographs can capture faces, premises, or identifying details, and warning records can contain officer contact information. Purpose, access, retention, and deletion must be defined for these data categories. The common architecture should store only information needed for the function and should avoid duplicating sensitive details in every service record.

Usability requires consistent naming, readable mobile content, visible progress, and recoverable errors. The field-support thesis reports a responsive layout overflow in the web target [8]. This is a material integration finding because an inaccessible result can make a functioning backend unusable. Accessibility evaluation should

include readable text, adequate contrast, screen-reader labels where supported, and risk communication that does not depend solely on color.

Maintainability depends on versioned schemas, modular adapters, and reproducible artefact loading. A model update may change class order or feature expectations even when a route name stays the same. Release review should therefore check model, feature schema, SHAP configuration, and mobile presentation together. Scalability should be assessed only after runtime memory and concurrency are measured; multiple workers can duplicate large model memory.

4.4 Integrated Workflow and Exception Handling

The normal prediction workflow validates the requested period, builds the ordered feature vector, executes the selected model, and formats the response. The same context is passed to the explanation layer. Before display, the client confirms that the response still belongs to the active request. This prevents a slow response for an earlier week from replacing the user's newer selection. A version mismatch or missing feature should block explanation rather than generate an attribution from a reconstructed input.

The intended intervention workflow requires valid area identity and provenance before constructing a decision state. Unsupported states and missing risk data should return a controlled unavailable result. A High-risk condition enables review of the recommended action and recipient. User cancellation terminates the warning attempt. Confirmation creates an identifiable transaction whose submission and storage outcomes are tracked independently. Figure 4.1 summarizes these exception boundaries.

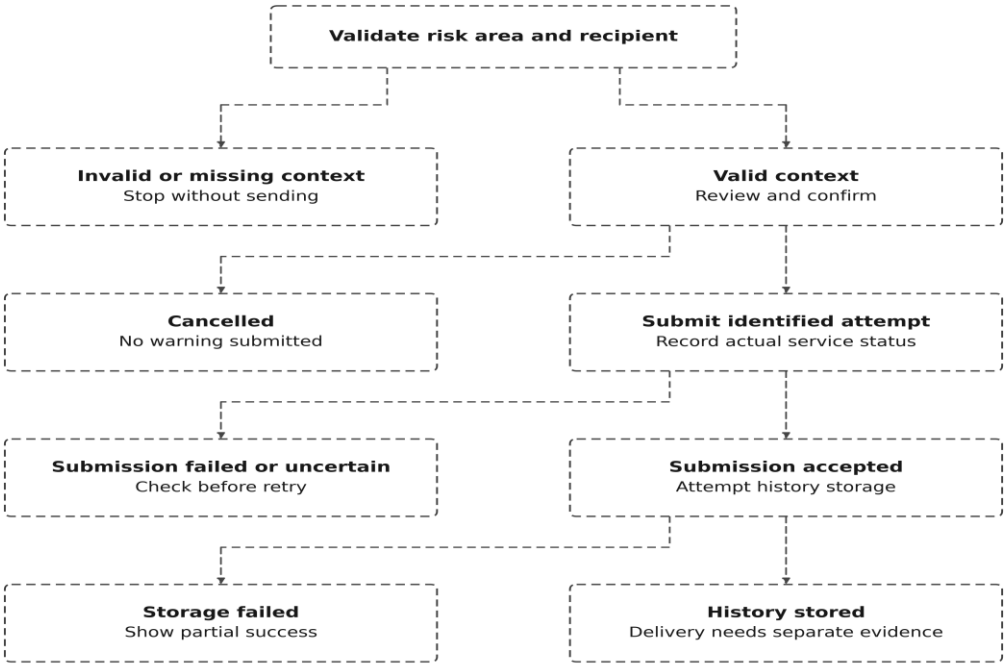


Figure 4.1 Intended warning transaction states and partial failure

Recipient mapping is a consequential dependency. An MOH area name can be ambiguous or inconsistently spelled, so the warning service should use stable identifiers rather than free text alone. Missing or disputed recipient mapping must prevent transmission. A retry should use an attempt identifier or another verified duplicate-control mechanism, because a timeout can occur after the service has already accepted a warning.

The image workflow distinguishes empty selection, invalid file, failed decoding, detector unavailability, and a valid result with no configured object. The last state is not a safety certificate for the premise. The officer should be able to continue physical inspection and record judgement without being forced to accept the model's output. Any later storage of manual corrections should preserve their provenance and should not silently become a training dataset.

The text workflow must respect routing outcomes. Clarification can be a correct result. A current-data request may need an unavailable response if live verified data are not connected. NLI review or insufficient evidence may also justify withholding a definitive answer. The severity model's weak overall holdout performance makes it

unsuitable as a sole gate for urgency decisions. Production behaviour should be reviewed by qualified public-health or clinical experts before relying on that signal.

4.5 Team Contribution and Dependency Matrix

The group has four distinct research owners. Table 4.3 records component responsibilities supported by the individual theses. Shared integration is a group outcome, but specific commit-level shares, meeting attendance, and effort proportions are not quantified. Appendix B provides a contribution declaration for member confirmation without assigning unsupported percentages.

Table 4.3 Research responsibilities and shared dependencies

Member and student ID	Research and development responsibility	Shared dependency
Keenavinna K.A.A.I.U IT22214034	Weekly data engineering Random Forest prediction and prediction API	Feature contract with C2 and risk provenance with C3
Hettiarachchi A.T IT22165848	SHAP analysis explanation formatting and prediction interpretation	Exact C1 model and mobile presentation
Wickramarachchi T.T. IT22243980	Q-learning intervention design and PHI warning workflow	Valid area context recipient mapping and history service
Adithya A A H IT22281982	PHI image workflow multilingual NLP and related backend integration	Shared Flutter shell authorization and deployment resources

Joint responsibilities include reviewing interface meaning, maintaining consistent application navigation, deciding authorization boundaries, and assembling integrated evidence. These tasks require agreement because a defect at a boundary can involve more than one component. For example, a feature mismatch may originate in prediction input preparation, explanation adaptation, or response formatting.

The source documents establish the component allocation but do not provide a consolidated planner export, repository history, or signed contribution record. The common report therefore treats the matrix as a traceable description of research scope. It does not claim a verified allocation of every integration task. Members should confirm the declaration and associate shared work with dated records before final submission.

The contribution structure also supports future maintenance. Each owner can document assumptions and test fixtures for their boundary while the group maintains a common release manifest. This reduces the chance that a later model update changes a shared behaviour without review by the dependent component owner.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

The common implementation context is a Flutter and Dart application communicating with Python-based backend services. FastAPI provides the prediction and health-intelligence interfaces described in the theses. The machine learning and XAI context uses Pandas, NumPy, scikit-learn, Joblib, and SHAP, while the field and language component uses PyTorch, transformer tooling, and an Ultralytics-compatible detector [2], [5], [8]. Table 5.1 identifies the shared layers and the limits of the implementation evidence.

Table 5.1 Technology layers and implementation evidence

Layer	Technology context	Reported role
Application	Flutter and Dart	Shared mobile interface and web development target
Tabular analysis	Python Pandas NumPy scikit-learn	Weekly preparation and Random Forest inference
Model explanation	SHAP and Joblib-based model loading	Global and local interpretation of prediction context
API	FastAPI and Uvicorn	Validation request handling and structured responses
Vision and language	PyTorch transformers detector tooling	Image detection and modular NLP inference
Authentication and data	Supabase	Officer access and data-service context
Packaging	Docker	Backend packaging described by C4
Collaboration	Git and GitHub	Version-control tools described in component work

The intervention thesis identifies Flutter, backend APIs, and Supabase history as its intended environment, while leaving the backend framework and email provider unconfirmed [6]. The common architecture can accommodate that component, but it does not assign an unverified email service or claim that C3 uses the exact deployed C1 environment. A final release specification must resolve this before describing the complete stack as one implementation.

The choice of a backend inference layer keeps heavy model processing outside the mobile application. This permits a common mobile experience without distributing large NLP checkpoints to each device. It also introduces network and hosting dependencies. A model that performs well offline may still be unsuitable for a user task if the server is slow, unavailable, or unable to load the artefacts.

Table 5.2 records the model-size profile described in C4 [8]. These are approximate artefact sizes, not measured peak memory or a complete storage budget. Tokenizers, dependencies, embeddings, intermediate tensors, and concurrent requests add further resource demand.

Table 5.2 Recorded field and language artefact sizes

Artefact	Primary task	Approximate size
clinical_nli_v1	Clinical consistency checking	531.81 MB
context_classifier_v1	Context routing	1060.69 MB
qa_rag_retriever_v1	Semantic retrieval	1060.67 MB
severity_classifier_v2	Urgency signal	1060.69 MB
vision_detector_v1	Environmental object detection	5.22 MB

The difference between the compact detector and the large language stack suggests a separable deployment strategy. A prediction service and an image service may have different infrastructure needs from an NLP service. Selective loading, shared encoders, distillation, or quantization are possible future directions, but their impact on accuracy, latency, and startup reliability must be measured. Model size alone does not identify the best deployment choice.

5.2 Development and Integration Process

The documented component work follows iterative development. C1 describes dataset preparation, feature engineering, model development, future-year evaluation, and API/mobile integration [2]. C2 describes feature mapping, SHAP processing, output formatting, and application presentation [5]. C4 describes isolated endpoint checks followed by shared Flutter screens, device communication, and responsive UI testing [8]. C3 presents an evidence plan for state/action verification, warning tests, and history validation [6].

These activities can be organized into a common sequence: establish a boundary, create the component artefact, verify its local behaviour, connect its interface, test the shared task, and record limitations. The sequence is a synthesis of the research process rather than a claim that a particular sprint board or continuous integration system was completed. No common sprint count, review frequency, or automated build coverage is quantified.

The most important interface-first work is agreeing on meaning before coding a dependency. C1 and C2 need the same ordered feature schema and class mapping. C1 and C3 need agreement on district estimates and area context. C3 and C4 need agreement on officer authorization and the information displayed during a professional workflow. Those agreements reduce the risk of technically valid requests producing semantically incompatible outputs.

Version control provides a mechanism for tracking changes, but merely naming Git does not establish reproducibility. A research release should identify a source commit, dataset version, model checksum, configuration, and test record. The application and backend must also identify which endpoints and schema versions they expect. A model file updated without its explanation adapter or class labels can invalidate the shared workflow while leaving the API reachable.

Reported development problems are informative. Physical device networking, Android configuration, responsive layout overflow, backend reachability, and model memory constraints appeared in the field-support integration context [8]. These show that group execution involves more than combining notebooks. Each defect should be associated with its layer and resolution evidence so that a later failure can be diagnosed without repeating the entire model experiment.

5.3 Deployment and Operational Configuration

C1 reports a FastAPI prediction service hosted through Hugging Face Spaces, with the Flutter application communicating through HTTP [2]. C2 uses the same backend–mobile pattern for prediction-related explanation [5]. C4 reports local backend operation, Android connection through development port reversal, Docker

configuration, and an unresolved full-stack hosting constraint [8]. These descriptions establish several environments, not a certified single production deployment.

Operational configuration must ensure that the application can reach the correct service and that the service can load the evaluated artefacts. A container starting successfully is insufficient if its model files are absent. Configuration should identify model locations, dataset locations, permitted client origins where relevant, service addresses, and authentication settings. Secrets should be supplied through protected runtime configuration rather than embedded in the application or committed to source control.

Readiness checks should distinguish process availability from model readiness. The prediction service needs its trained classifier and required historical records. The explanation service needs a compatible model and schema. The detector needs its checkpoint and image-decoding dependencies. The NLP service needs classifiers, evidence passages, embedding index, and consistent metadata. The intervention service additionally needs a confirmed state/action configuration and recipient mapping.

Monitoring should record request identifiers, service version, processing duration, controlled error categories, and resource failures. Sensitive user content should not be retained merely to make debugging convenient. Model and corpus identifiers are particularly useful because a change in either can alter behaviour. A response-quality concern should be traceable to the exact checkpoint and evidence version that produced it.

Backups and restoration are operational requirements rather than demonstrated outcomes. Data and warning history should have an agreed backup policy, access protection, retention period, and tested recovery procedure. Model artefacts and release manifests need separate preservation because restoring a database does not restore the inference environment. A common recovery test should restore a known release and execute at least one prediction/explanation request and one restricted workflow.

No measured uptime, recovery time, load limit, or complete production backup test is established. The current deployment conclusion is limited to the reported prediction

service and development integration evidence. Full service readiness requires memory and latency measurement, protected configuration, repeatable deployment, and verification of the remaining intervention path.

5.4 Evidence of Team Execution

The individual theses provide component scope, method descriptions, implementation narratives, and selected results. They are the current evidence for team allocation and research progress [2], [5], [6], [8]. Table 5.3 identifies what each evidence category supports and what it cannot establish. This distinction prevents an implementation narrative from being treated as a substitute for a dated integrated test run.

Table 5.3 Execution evidence and interpretation

Evidence category	Supports	Does not establish alone
Individual thesis	Component scope method and reported results	Exact common release or all shared tasks
Experiment screenshot	Visible metric and recorded task output	Full dataset independence or reproduction
API response example	A documented request path returned output	Complete regression coverage or field accuracy
Shared application narrative	Navigation and integration context	Full four-component acceptance
Version-control tool description	Intended change-tracking mechanism	Commit history review or contribution percentages
Prospective test matrix	Defined validation work	Executed test outcomes

The current report does not claim consolidated MS Planner, Teams, meeting-minute, or repository-history evidence. Those records are useful for substantiating shared execution and should be associated with contribution declarations if available. Their absence does not remove the component work, but it limits conclusions about exact collaboration history and reproducible release control.

The strongest immediate execution evidence is the measured prediction experiment and the recorded NLP outputs, together with reported application and backend integration. The weaker areas are C3 implementation verification, full vision evaluation, and a single group acceptance record. This unevenness guides the next

work: consolidate the artefacts and test their boundaries before claiming complete readiness.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

The evaluation framework separates four questions. First, does each model perform its defined task on an appropriate held-out dataset? Second, do service interfaces preserve the meaning and identity of their inputs and outputs? Third, can a user complete an integrated task under realistic conditions? Fourth, are the system's uncertainty, limitations, and consequential decisions handled responsibly? None of these questions can be answered by one combined accuracy score.

For C1, the recorded experimental design is temporal: train on 2022–2024 and test on 2025 [2]. Accuracy and weighted F1 are the reported measures. A fuller evaluation should include class-level precision and recall, confusion entries, target thresholds, exact test count, and multiple future periods. Those additions would clarify whether rare High-risk observations are missed even when aggregate performance appears useful.

For C2, acceptance requires correspondence between the explained output and the displayed prediction. Tests should verify feature order, names, values, class mapping, model version, and attribution reconstruction in the configured output space. Explanation latency and user understanding are separate measures. The supporting prediction accuracy in C2 does not quantify explanation faithfulness or usability [5].

For C3, functional success requires a defined state, available action, documented policy, traceable reward update where learning occurs, and accurate warning status. A policy comparison must use the same states and evaluation conditions as a rule-based baseline. Expert approval, simulation reward, communication success, and measured disease outcomes are different constructs and must be evaluated separately [6].

For C4, classification uses accuracy and macro F1; retrieval uses top-k recall and mean reciprocal rank (MRR); NLI also uses coverage and accuracy on non-neutral decisions; detection requires class-level precision, recall, and mean average precision (mAP). The component's existing NLP evaluations provide several of these measures, whereas the detector benchmark remains unreported [8].

Table 6.1 defines group acceptance evidence. The criteria are testable requirements for the complete prototype rather than retrospective numerical targets. Appendix C expands the prospective cases and clearly marks them as unexecuted unless supported by a component record.

Table 6.1 Group evaluation questions and acceptance evidence

Question	Required evidence	Current scope
Prediction validity	Temporal split metrics class diagnostics	Aggregate C1 result reported
Explanation consistency	Same input model class and output reconstruction	Integration described no common numeric check
Action validity	Defined states actions rewards and policy trace	C3 preliminary design
Workflow reliability	Repeated end-to-end and failure-path records	Selected component paths reported
Health information grounding	Retrieval NLI and expert answer review	Model metrics and Sinhala response example
Operational readiness	Resource load recovery and stakeholder tasks	Resource constraint identified no full benchmark

The report does not infer confidence intervals from undocumented denominators or calculate a system pass percentage from incomplete test counts. Evidence is presented at the level actually established, with the dataset and task context attached to each result.

6.2 Integration and End to End Testing

The prediction thesis describes testing of data structure, feature generation, temporal ordering, API handling, mobile display, and the request-to-result workflow [2]. It reports that prediction functionality was accessible through the FastAPI service and Flutter application. However, the common record does not include an executed consolidated matrix with exact test inputs, outputs, dates, and pass/fail counts. These statements support reported technical integration, not a calculated group regression success rate.

The XAI thesis describes functional, interface, input-validation, and explanation-consistency testing [5]. Its key requirement is the link between prediction and attribution. An integrated test should retain both the original model input and

returned explanation so that the mapping can be inspected. A successful chart display does not prove that a feature index or class index was correct. This boundary needs a repeatable numerical and semantic test in the final release.

C4 provides a specific backend example. The /v1/analyze route returned HTTP 200 for a Sinhala question-answering path, with language metadata, a sufficient-context result, answer text, and evidence items containing identifiers, source URLs, and scores [8]. This is direct evidence of orchestration and serialization for the recorded request. It is not a complete multilingual answer-quality study, nor does HTTP 200 prove that every statement in the response was medically appropriate.

The C4 narrative also describes an authorized officer gate, an image inference path, Android communication, and web testing. It identifies responsive layout overflow and infrastructure limitations. These observations show why integration testing must include the interface and runtime, rather than only checking model files. Exact authorization coverage, image test count, and latency values remain unreported.

The C3 test plan is prospective [6]. Valid recommendations, High-risk triggers, reward handling, correct recipients, warning content, history retrieval, invalid input, email failure, and database failure are defined cases rather than measured successes. The common report therefore marks the full risk-to-warning scenario as pending implementation and execution evidence. No warning is counted as delivered and no Q-learning improvement is inferred.

The principal group test is a complete professional scenario: review a valid risk context, inspect a consistent explanation where applicable, review a proposed action for a valid area state, confirm a mapped recipient, observe truthful warning status, and verify history. That scenario must also be repeated with stale data, missing area mapping, unavailable models, cancelled confirmation, and partial storage failure. A successful happy path without these cases would provide limited assurance.

6.3 Overall Results

6.3.1 Prediction Results and Model Provenance

Table 6.2 records the two distinct prediction experiments described in the component theses. The C1 result is the primary reported result for the current district prediction component. The C2 result is retained as a separate supporting experiment because its training period and application unit differ. Neither record is relabelled as SHAP accuracy or as the overall system accuracy.

Table 6.2 Prediction experiment records retained separately

Record	Training and evaluation context	Reported metrics
C1 prediction thesis [2]	2022–2024 training 2025 unseen test district model	Accuracy 84.31% weighted F1 82.11%
C2 XAI thesis [5]	2015–2024 training independent 2025 test supporting model	Accuracy 86.35% precision 86.37% recall 86.35% F1 86.36%
C2 later application [5]	2026 Weeks 1–22 220 area-week observations	143 Medium 54 Low 23 High predictions

The discrepancy is material to integration. Different training records, feature sets, target thresholds, model checkpoints, or evaluation units could explain different metrics, but the exact explanation is not established. Selecting the larger number would obscure that uncertainty. A common release needs a model manifest linking each reported experiment to its dataset, feature schema, target rules, geographic unit, and checksum.

The C2 area-week application produced a distribution of predicted labels: 65.00% Medium, 24.55% Low, and 10.45% High [5]. These are output frequencies, not accuracy measurements. Without corresponding observed labels and an appropriate area-level evaluation, they cannot establish geographical predictive validity or the correctness of recommendations derived from those labels.

The C1 future-year result indicates useful classification within its defined temporal setting. It does not establish performance across all later years or all MOH areas. Weighted F1 also weights classes by their frequency, so a common evaluation should examine High-risk recall directly. The current aggregate result is promising for the research prototype, while class-level and multi-period validation remain necessary.

6.3.2 Explanation Results

The XAI thesis reports global and local SHAP integration and application presentation [5]. Its technical contribution is linking feature contributions to prediction context. No final common feature-ranking table, local attribution vector, reconstruction tolerance, or explanation-specific user score is established here. The report therefore does not invent a most influential feature or quantify an improvement in user trust.

Global and local analysis have different interpretations. A global mean absolute attribution ranks contribution magnitude across a specified observation set. A local attribution describes one class output for one feature vector. The dataset used for the global summary and the configured output space matter. A positive contribution should be read in that model context rather than as an unconditional statement that the feature increases real-world dengue transmission.

The integrated finding is that interpretation can be delivered through the same backend and mobile pathway as prediction. The main unresolved evidence is whether the final common release preserves exact model/input identity and whether users understand the explanation. These issues are testable and should be evaluated independently from the predictive benchmark.

6.3.3 Multilingual Classification and Retrieval Results

Table 6.3 synthesizes the recorded NLP results from C4 [8]. Scores are reported as decimals to retain the source representation. The tasks, denominators, and test types differ, so they are not averaged.

Table 6.3 Recorded NLP task results

Task and evaluation	Dataset context	Reported result
Context held-out test	204 records	Accuracy 1.0000 macro F1 1.0000
Context dialogue stress	Separate multi-turn challenge	Accuracy 0.9167 macro F1 0.8939
Severity official holdout	32 cases	Accuracy 0.4688 macro F1 0.4618
Emergency Alert class	Within severity holdout	Recall 0.8750 precision 0.6364

Closed-corpus retrieval	189 queries 507 passages	Top-1 0.5132 top-3 0.6614 top-5 0.7196 MRR 0.6130
Zero-shot clinical NLI	183 claim/evidence pairs	Accuracy 0.7650 macro F1 0.8453
Clinical NLI decisions	Coverage 0.8251	Accuracy when decided 0.9272
Misinformation held-out test	408 balanced test records	Accuracy 0.9951 macro F1 0.9951
Misinformation hard challenge	Separate hard examples	Accuracy 0.8333 macro F1 0.8286
Numeric claim verification	1442 labelled claims	All classified correctly in recorded run

The controlled context score is strong, but the dialogue stress result is lower. This demonstrates the difference between a standard split and conversational ambiguity. A short follow-up may depend on prior turns or may lack enough information to select a route. Context classification should therefore retain clarification and review paths rather than assume perfect performance in open use.

Retrieval performance improves when more passages are considered: top-1 exact-answer accuracy is 0.5132, with top-3 and top-5 exact-answer recall of 0.6614 and 0.7196. These results support the use of multiple evidence candidates. They also show that a substantial portion of test queries did not recover the exact answer even at top 5. A downstream consistency model cannot guarantee recovery when the relevant evidence is absent.

The NLI record contains 32 neutral or needs-review outcomes among 183 pairs, with decision coverage of 0.8251. Accuracy of 0.9272 applies only when a non-neutral decision was made; overall accuracy is 0.7650 [8]. Reporting only the conditional accuracy would overstate performance across all inputs. The review state is a useful mechanism for uncertainty, but its operational handling still needs answer-level evaluation.

The misinformation classifier's balanced held-out confusion record contains 202 correctly identified Supported examples, two Supported examples labelled Misinformation, and all 204 Misinformation examples identified correctly. Its hard-example result is lower. The numeric checker correctly classified 1017 Misinformation and 425 Supported claims in its recorded run. These are bounded

tests of the data and claim formats, not general verification of every possible health statement or live surveillance count.

6.3.4 Severity and Field-Support Limitations

The severity holdout is the most consequential weak result. Overall accuracy is 0.4688 and macro F1 is 0.4618 on 32 cases. Emergency Alert recall is 0.8750, while recalls for Low Concern, Moderate Concern, and High Concern are 0.3750, 0.3750, and 0.2500 respectively [8]. Emergency recall is relevant, but it does not make the multiclass classifier reliable enough for autonomous urgency decisions.

The 32-case holdout is small and differs from the larger development dataset. The source describes 1232 severity records, balanced across four classes, with a 960/120/152 development split before the later holdout [8]. The holdout finding should be retained rather than replaced by an optimistic development result. Future work must examine label quality, domain shift, language distribution, and expert review.

The vision pathway is reported as implemented, but detector precision, recall, mAP, inference latency, class inventory, and final test image count are unreported. There is therefore no basis for a numeric claim about inspection accuracy. An implemented upload-to-detection path establishes software capability. Its practical value requires independent images, class-level assessment, and PHI review under real field conditions.

6.3.5 Integrated Maturity and Deployment Findings

Table 6.4 summarizes the system at the level justified by the evidence. This is a maturity assessment of the research artefact, not a certification or a release approval.

Table 6.4 Integrated capability and evidence status

Capability	Current finding	Remaining evidence
District prediction	Model result and API/mobile integration reported	Unified model manifest and class diagnostics
SHAP explanation	Global local and application integration described	Numeric consistency and user comprehension
Intervention agent	State/action and warning design documented	Code policy reward and executed tests

PHI vision	Image inference path reported	Dataset inventory and detector benchmark
Multilingual health intelligence	Multiple task metrics and Sinhala API example	Answer-level expert and stakeholder validation
Shared application	Navigation authorization context and device work described	Full four-component regression matrix
Full cloud service	Prediction hosting reported NLP resources constrained	Complete load readiness and recovery tests

The overall result is a connected research architecture with implemented and measured subpaths, plus a defined action-oriented design that remains incomplete. No single complete-system metric is defensible. The strongest evidence lies in temporal prediction, curated NLP experiments, and selected technical integrations. The main blockers are model-provenance reconciliation, intervention verification, vision benchmarking, formal user evaluation, and infrastructure validation.

6.4 Stakeholder and User Validation

No formal group stakeholder study with participant counts, recruitment method, consent procedure, task results, and statistical analysis is established in the research record. The XAI thesis explicitly does not report a separate structured usability evaluation, while C4 leaves formal PHI and public-user findings outstanding [5], [8]. The intervention thesis likewise does not claim completed stakeholder validation [6].

Developer testing and interface presentation are useful technical evidence, but they do not establish that PHIs interpret outputs correctly or that public users understand answer limitations. A future evaluation should include realistic professional tasks, preferred language forms, recoverable failures, and explanation comprehension. Appendix D provides a prospective protocol that separates task observation from opinion ratings.

For XAI, participants should identify the selected period, risk class, and meaning of a contribution, and explain why an attribution is not causal proof. For intervention support, they should identify area provenance, review a proposed action, and distinguish warning submission from delivery. For inspection, they should recognize that missed objects require physical review. For health information, they should locate evidence and understand clarification or review outcomes.

Results should record successful completion, assistance, time, errors, misunderstanding, and qualitative concerns. A bespoke questionnaire should not be reported as a System Usability Scale score unless that standardized instrument and its scoring are actually used. No participant number or rating is invented in this report.

6.5 Achievement of Common Objectives

Table 6.5 maps the available evidence to the six group objectives. A partial status reflects substantive progress with unresolved group evidence. It is preferable to an unqualified completion claim where an interface or validation stage remains preliminary.

Table 6.5 Objective achievement against current evidence

Objective	Assessment	Justification
G1 Temporal risk pipeline	Substantially achieved at component level	C1 reports weekly data model and future-year metrics
G2 Consistent explanation	Partly achieved	SHAP integration described common version check outstanding
G3 Reviewed risk-to-action support	Design established implementation unverified	C3 defines framework but no measured learning or warning tests
G4 Field and multilingual support	Partly achieved	NLP metrics and paths reported vision benchmark outstanding
G5 Common application integration	Partly achieved	Shared shell and selected paths reported full C3 acceptance absent
G6 Responsible operational readiness	Partly achieved	Limits and resource needs identified field and cloud validation pending

The project meets an important research objective by defining a coherent division of work and identifying the contracts needed for combined operation. It also provides usable component evidence rather than only an architectural proposal. However, the common aim includes evaluation of the integrated prototype, so the missing four-component test and stakeholder record prevent a claim of full achievement.

The reported prediction result supports G1 within its district and temporal scope. It does not automatically complete G3, because action selection requires valid area context and a tested policy. Likewise, a strong misinformation result supports part of

G4 but does not complete G6, because deployment and comprehension are different outcomes. The objective mapping preserves those dependencies.

6.6 Integrated Limitations and Threats

The first limitation is model and dataset provenance. The two prediction accounts differ in training period, performance, and operational granularity. Until reconciled through actual artefacts, a common benchmark remains uncertain. This threatens reproduction and the validity of downstream explanations or recommendations that assume one model context.

Temporal generalization is bounded by one reported future test year for C1. Additional years, rolling evaluations, and drift monitoring would strengthen external validity. Future inputs derived from seasonal history have a different status from observed inputs. Accuracy on observed 2025 records should not be generalized to all future-week scenarios.

Area-level context is constrained by proxy data. The separate MOH dataset is not a complete official surveillance source [2]. C3's dependence on area risk makes this a system limitation rather than a local data note. An operational trial must document how area risk is obtained and whether the target and intervention state are valid for that unit.

The NLP evaluations are curated and differ by task. Perfect context classification and numeric checking on bounded sets do not eliminate language variation, adversarial phrasing, missing evidence, or guideline updates. The weak severity holdout is especially important. Answer-level correctness, faithfulness, and safe handling of uncertainty remain separate validation needs.

Vision external validity cannot be assessed without a final dataset and benchmark. Lighting, camera angle, scale, clutter, occlusion, and unseen objects may affect detections. An apparent environmental object does not establish breeding status, and no detection does not establish safety. A field study must evaluate how officers use the output rather than only how a detector matches labels.

Operational validity is limited by the missing consolidated test run, incomplete intervention evidence, and unvalidated full hosting configuration. Developer-led

testing can also create confirmation bias. Table 6.6 summarizes the principal threats and appropriate next evidence.

Table 6.6 Main validity threats and required evidence

Threat	Effect on conclusion	Next evidence
Different model accounts	Uncertain common prediction baseline	Checkpoints datasets schemas and experiment manifests
Single future holdout	Limited long-term generalization	Rolling multi-year external tests
Area proxy information	Overstated local precision	Official data or evaluated proxy mapping
Small severity holdout	Uncertain urgency behaviour	Expert-labelled independent challenge set
Missing vision benchmark	Unknown field detection quality	Independent multi-site image test
Missing stakeholder study	Unknown understanding and workflow value	Observed PHI and public-user tasks
Unverified C3 learning	No demonstrated policy improvement	Defined environment baseline and reward traces
Resource-constrained hosting	Uncertain availability and responsiveness	Runtime memory concurrency and recovery tests

The conclusions are therefore limited to prototype architecture, reported component performance, and selected technical paths. They do not establish field effectiveness, causal disease reduction, or universal generalization. These limits also define a focused research agenda for the next release.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value Proposition and Beneficiaries

The proposed value is a common environment for reviewing dengue risk, understanding model factors, considering actions, inspecting images, and accessing evidence-oriented information. For professional users, continuity may reduce repeated context entry and make outputs easier to review. For public users, local-language information may improve access. For the research team, modular interfaces provide a basis for extending and evaluating the artefact.

These benefits remain prospective at the institutional and population level. The current evidence establishes software and research capabilities, not a measured reduction in workload, improved adherence, willingness to pay, or avoided disease cases. Commercialization should therefore begin with an evaluation partnership rather than a claim of proven operational savings.

The National Dengue Control Unit identifies surveillance, vector control, risk communication, and coordinated response as related activities [14]. The integrated application is relevant to that workflow structure. It should complement approved institutional practice and should not imply an endorsed connection to official surveillance systems without an actual partnership and verified data access.

Beneficiaries have different priorities. A PHI needs practical image handling and reliable access. An MOH team needs valid local context and defensible action support. A programme manager needs traceability and maintenance. A public user needs understandable evidence and clear uncertainty. A sustainable product must prioritize these tasks rather than assuming that adding more models always adds value.

7.2 Market Alternatives and Differentiation

Relevant alternatives include existing surveillance workflows, spreadsheets and dashboards, standalone predictive models, manual intervention rules, field documentation tools, and general information search. These are functional categories

rather than a priced market survey. Table 7.1 compares their task coverage with the proposed integrated platform without claiming numerical superiority.

Table 7.1 Functional alternatives and proposed differentiation

Alternative category	Typical contribution	Additional proposed common capability
Surveillance dashboard	Summarizes reported conditions	Prediction and explanation linked to context
Standalone model	Produces an estimate	Mobile delivery and interpretation
Manual action rules	Transparent risk-to-action mapping	Research on feedback-based state-action selection
Field documentation tool	Captures observations	AI attention support with officer review
General health search	Finds information sources	Dengue routing and structured evidence metadata
Separate component apps	Provides individual functions	Shared navigation authorization and context

The most defensible differentiation is the integration of bounded functions. C1 supplies temporal prediction; C2 supplies explanation; C3 defines advisory action and warning support; C4 supplies observation and evidence-oriented language support. This creates a common workflow proposition, but its value must be tested as a whole. Four feature lists do not establish that users can complete a task more effectively.

A rule-based intervention approach is a useful comparison because it can be transparent and easy to audit. Q-learning should demonstrate an advantage under a defined environment before it becomes the preferred operational mechanism. Where reliable rewards are unavailable, professional rules may remain a stronger baseline. The research purpose of C3 is to investigate feedback-based selection, not to assume its superiority.

The current project also has costs associated with differentiation. Modular language models improve task-level analysis but increase memory and maintenance. Source-linked answers require corpus review. Shared navigation requires regression testing across components. These costs should be assessed alongside any user benefit rather than hidden behind a claim of an all-in-one system.

7.3 Business Model and Feasibility

A feasible initial model is an institutional research pilot with a defined partner, limited users, controlled data access, and a supported prototype release. This approach can evaluate task value before selecting a long-term commercial arrangement. Possible later arrangements include a maintained institutional service or a research-supported open implementation, subject to ownership and licensing review. No contract, customer commitment, funding award, or revenue is claimed.

Key resources are the mobile application, trained and verified artefacts, data preparation pipeline, reviewed health corpus, backend infrastructure, and technical support. Key activities are data updates, model evaluation, source review, user training, access administration, incident handling, and release maintenance. These activities continue after development and should be included in operational planning.

The cost structure is dominated by more than initial software creation. Memory-intensive NLP hosting, model storage, monitoring, backups, language review, field data collection, and officer support all require resources. The present record provides model file sizes but no validated service capacity or financial estimate. It therefore does not justify a subscription price, cost-per-user figure, or profit forecast.

Technical feasibility is strongest for the reported prediction service and local integration paths. Full-stack feasibility remains limited by resource demand and unverified C3 behaviour. A staged pilot can begin with independently valid functions while keeping incomplete capabilities unavailable or clearly experimental. It should not advertise a complete action-optimization service before the policy and warning path are tested.

Operational feasibility depends on institutional fit. The platform needs a responsible maintainer, a source-update process, a help route, and rules for responding to incorrect outputs. Users need training in interpreting probabilities, SHAP contributions, proxy context, and warning status. These obligations are part of the product rather than optional documentation added after launch.

7.4 Adoption Intellectual Property and Compliance

Adoption barriers include trust, connectivity, workflow duplication, unclear geographical precision, inconsistent language performance, and maintenance capacity. A pilot should address these through observed tasks and feedback rather than assuming that technical integration will produce institutional adoption. The most immediate barrier is evidence completeness: the common release needs a verified model manifest and tested intervention path.

Ownership and licensing must be reviewed across code, datasets, model checkpoints, and source passages. The student's original work, third-party libraries, pretrained models, public guidance, and contributed field images may carry different terms. Public accessibility does not automatically authorize redistribution or commercial model training. A release inventory should record the source and permitted use of each asset.

Responsible operation also requires data governance. Officer identity, contact mappings, warning history, and field images should have a defined purpose, restricted access, retention policy, and deletion process. Access to official surveillance systems should occur through an authorized arrangement. The report does not assert legal compliance certification or provide a current legal opinion; deployment must be reviewed under the applicable institutional and legal requirements.

WHO's health-AI guidance emphasizes ethical governance and accountability for systems used in health settings [15]. The project applies that framing through advisory outputs and human review. In practice, this requires truthful claims, visible uncertainty, traceability, and an accountable person or institution for maintenance. A label stating decision support does not compensate for a misleading interface or unreviewed high-risk recommendation.

The user interface should preserve boundaries. Prediction is an estimate. SHAP explains a model. Image detection identifies a configured object. NLP retrieval supplies candidate evidence. Q-learning evaluates actions according to its defined reward. None should be presented as an independent clinical or statutory authority.

Human confirmation should occur before warnings and consequential actions, with a record of what was actually accepted and transmitted.

Academic integrity also applies to this report. Generative AI assisted consolidation, language, structure, and document preparation. It did not create new experimental data or establish missing test outcomes. Appendix E records that assistance and the authors' responsibility for technical verification, source checking, contribution confirmation, and final institutional disclosure.

7.5 Sustainable Development Goal Contribution

The project has a plausible pathway to supporting early warning and organized dengue prevention, consistent with the Goal 3 alignment identified in Chapter 1 [9]. That pathway passes through several intermediate steps: valid data, interpretable estimates, appropriate action review, reliable field support, understandable information, and sustained institutional use. A failure at any step weakens the claimed social value.

The current evaluation supports some technical steps but does not measure population impact. Future evaluation should first assess whether users complete tasks accurately and understand limitations. A later operational study could measure response timeliness, inspection documentation quality, and appropriate use of information. Claims about disease outcomes would require longer follow-up and a design that considers reporting changes, seasonal conditions, and concurrent public-health activities.

Sustainability includes infrastructure and maintenance. Large models create ongoing compute demand. Corpus review requires people with language and domain expertise. Hardware and connectivity constraints can exclude some users. A lighter model is beneficial only if it retains adequate quality; a low-cost service is beneficial only if it remains available for the intended task. Resource optimization should therefore be assessed together with reliability and language performance.

Potential negative effects include false reassurance, unnecessary warnings, misinterpretation of proxy area data, and unequal information quality across languages. A responsible pilot should measure these risks and revise the interface or

disable weak functions when evidence is insufficient. The weak severity benchmark is a concrete reason to prioritize careful routing and expert review.

The project's sustainable contribution is thus conditional. It provides a research foundation for integrated support and identifies the evidence needed to make that support dependable. Continued validation, institutional partnership, and maintenance are necessary before a claim of sustained public-health benefit can be made.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

The common problem is addressed through a coherent architecture connecting risk prediction, explanation, advisory action planning, field observation, and multilingual information. The four components have complementary responsibilities and can share a mobile interface and backend conventions without losing their distinct analytical roles. This is a meaningful system contribution because the value of dengue decision support depends on the connection between outputs and the tasks users must perform.

The research evidence is uneven but informative. C1 reports a district-level Random Forest model evaluated on an unseen 2025 period, with 84.31% accuracy and 82.11% weighted F1 [2]. C2 describes SHAP-based global and local interpretation and application integration [5]. C4 supplies multiple NLP experiments, a recorded Sinhala API response, and field-support integration context [8]. C3 establishes a Q-learning and PHI-warning design but does not report measured learning or completed warning validation [6].

These findings support a prototype-level conclusion. The project has developed and documented several connected capabilities and defined the remaining action-oriented pathway. It has not yet demonstrated full four-component operational effectiveness. Different prediction experiment records, incomplete vision metrics, absent formal stakeholder results, and resource-constrained cloud operation limit the common evaluation.

The results also show why a single system accuracy would be misleading. Prediction, explanation, retrieval, urgency classification, warning transmission, and usability measure different constructs. Strong performance on one task does not remove a weak dependency. The severity holdout, for example, remains weak despite strong curated context and misinformation results. The architecture must therefore support bounded outputs and truthful uncertainty.

The overall aim is partly achieved at the group level. The common research framework, component allocation, shared implementation context, and measured

subpaths are established. Full achievement requires a verified release manifest, valid area-to-intervention mapping, complete C3 tests, vision benchmarking, stakeholder evaluation, and deployment assessment. No claim of causal dengue reduction or autonomous public-health optimization is justified by the present evidence.

8.2 Principal Integrated Contribution

The principal contribution is the organization of multiple AI methods around one accountable user workflow. Historical records generate a risk estimate; SHAP provides model interpretation; a proposed action layer connects valid area context to reviewed recommendations; and field and language services support observation and information access. Shared context and evidence conventions provide continuity across those functions.

A second contribution is the explicit preservation of scale and provenance. District risk, area proxy context, premise observations, and query-level health information remain separate. This prevents an interface from implying a precision or authority that the underlying research does not establish. The same principle applies to model identity: the prediction and explanation must remain connected to the exact evaluated artefact.

A third contribution is a layered evaluation model. The report separates offline scores, API examples, mobile integration, human understanding, and operational outcomes. It records preliminary design as preliminary design and measured results as measured results. This creates a stronger basis for the next study than treating all available screenshots or narratives as proof of complete readiness.

The common interfaces, contribution matrix, prospective tests, and validation protocol can guide subsequent implementation. They do not replace execution evidence, but they define what evidence is needed and which component owns the relevant boundary. This supports a maintainable continuation of the work beyond the individual theses.

8.3 Future Directions

The first priority is reconciling the prediction artefacts. The team should identify the selected checkpoint, training records, feature order, target rules, geographical unit, and evaluation script. The 84.31% and 86.35% accounts must be linked to separate manifests or corrected where one is an outdated description. The SHAP adapter should then be tested against the selected release with a repeatable attribution consistency check.

The second priority is completing C3. This requires exact state encoding, action labels, reward meaning, update timing, learning parameters, and policy behaviour. A rule-based baseline should be evaluated under equivalent conditions. The warning path needs controlled tests for valid and missing recipients, confirmation and cancellation, timeout and retry, transmission status, and independent history storage. No real disease benefit should be inferred from computational reward alone.

The third priority is strengthening field and language evaluation. The detector needs a final image inventory, permission and annotation record, independent split, class-level benchmark, and realistic PHI task study. NLP needs expert-reviewed answer evaluation, additional language and code-mixing cases, and a stronger urgency challenge set. Tamil support should be claimed only after implementation and separate validation.

The fourth priority is a consolidated group acceptance run. The same release should execute prediction/explanation, field inspection, text analysis, and the validated action workflow. It should include authorization failures, stale or incomplete data, unsupported area context, absent model artefacts, and partial transaction failures. Exact inputs, outputs, versions, outcomes, and evidence references should be retained so that the group result is reproducible.

The fifth priority is realistic deployment and adoption. Measure peak memory, startup time, tail latency, concurrency, and recovery before selecting an operating configuration. Evaluate model routing or optimization against quality, not only size. Conduct a limited institutional pilot with trained users and a defined maintainer. The

pilot should assess comprehension, task value, failure handling, and appropriate use before broader dissemination.

These priorities move the project from a connected research prototype toward an evaluated operational tool. The existing architecture and component evidence provide a basis for that work, while the next release must resolve the dependencies that currently limit a complete system claim.

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APPENDICES

Appendix A Integrated Interface Specification

The specification in Table A.1 records logical information requirements for a common release. It does not assert actual database columns or exact request field names. The final API schema must be checked against the selected source and running service. The documented route inventory appears in Table 4.2.

Table A.1 Logical interface records and validation rules

Record	Required information	Validation rule
Prediction	Period unit input origin features schema model class probabilities	Inputs match selected model and probabilities retain class mapping
Explanation	Prediction reference class output space contributions version	Same model and feature vector as associated prediction
Area context	MOH identifier period source proxy flag completeness	District information is not relabelled as area validation
Recommendation	State policy action labels context reference	Valid state and approved action catalogue required
Warning attempt	Confirmed area recipient actions timestamp attempt status	Missing recipient or cancellation prevents submission
Warning history	Attempt reference retained content submission storage status	Transmission and persistence outcomes remain separate
Inspection	Officer context detector labels scores coordinates	Valid image and authorized request before inference
Text analysis	Language route context evidence sources and review status	Evidence and uncertainty are retained in presentation

The prediction–explanation contract should be immutable for one request. If a later model version is selected, the original explanation should remain associated with its original prediction. A regenerated explanation is a new result with its own version. This supports comparison without silently changing historical interpretations.

For warning transactions, a unique attempt identity is a proposed reliability requirement. The implementation must define how it prevents duplicate submission after a timeout. The record should retain the recommendation as reviewed at the time

of confirmation rather than recalculating it later under a changed policy. No specific database mechanism is assumed.

The source example from /v1/analyze establishes a route, language, context result, answer, and evidence metadata [8]. The wider logical contract adds version and request identity requirements for common traceability. These additional requirements remain release design work until verified in implementation.

Appendix B Team Contribution Declaration

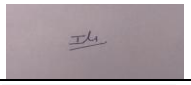
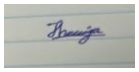
The members' research responsibilities are recorded in Table B.1 using the names and student identifiers in the individual theses. Signatures confirm the allocation and shared contribution account after member review. No effort percentages or commit totals are assigned without evidence.

Table B.1 Component allocation for member confirmation

Member	Student ID	Component
Keenavinna K.A.A.I.U	IT22214034	Data engineering and machine learning prediction
Hettiarachchi A.T	IT22165848	SHAP-based explainable artificial intelligence
Wickramarachchi T.T.	IT22243980	Dengue intervention optimization agent
Adithya A A H	IT22281982	PHI inspection and multilingual health intelligence

We confirm that the component responsibilities above reflect the scope of our individual theses. Shared outputs described in this common report include the integrated problem definition, architecture, interface requirements, evaluation synthesis, and future validation plan. Specific implementation contributions should be read with the evidence status in Chapters 5 and 6. Our confirmation does not convert planned tests or unverified metrics into completed outcomes.

Table B.2 Member confirmation signatures

Student ID	Signature	Date
IT22214034		2026.10.11
IT22165848		2026.10.11
IT22243980		2026.10.11
IT22281982		2026.10.11

Appendix C Integrated Test Matrix

Table C.1 defines prospective common-release tests. The status identifies the existing evidence boundary; it is not an executed test result. A future execution record must add exact input, expected output, observed output, release identity, date, and evidence reference.

Table C.1 Prospective group acceptance and exception tests

ID	Scenario and expected behaviour	Current evidence status
GT01	Valid period produces district prediction with context	Component workflow reported common run pending
GT02	Same prediction yields correctly mapped SHAP output	Integration described numerical common test pending
GT03	Model or feature mismatch blocks explanation	Required failure test not executed in common record
GT04	Future request identifies seasonal input assumptions	Mechanism reported presentation check pending
GT05	Proxy area context remains visibly distinct	Source distinction documented common UI check pending
GT06	Invalid area state prevents recommendation	C3 design test pending
GT07	Action selection matches defined policy	C3 implementation verification pending
GT08	Reward update matches documented transition	C3 learning trace pending
GT09	Missing recipient prevents warning submission	C3 design test pending
GT10	User cancellation ends warning attempt	C3 design test pending
GT11	Confirmed warning records truthful submission status	C3 communication test pending
GT12	Storage failure preserves partial-success distinction	C3 failure test pending
GT13	Retry does not duplicate accepted warning	Duplicate-control design and test pending
GT14	Valid image returns structured detections	C4 component path reported common run pending
GT15	Corrupt or oversized image is rejected safely	Requirement described common run pending
GT16	No detection avoids a premise-safe claim	Wording and field-task check pending
GT17	Restricted endpoint rejects unauthorized request	Auth context described server test pending
GT18	Sinhala query returns evidence metadata	Recorded C4 API example common rerun pending

GT19	English and Singlish queries retain meaning	Dataset evidence reported answer-level test pending
GT20	Current-data query avoids stale factual assurance	Routing class evidenced end-to-end check pending
GT21	Insufficient evidence returns review or clarification	Pipeline design described common run pending
GT22	Unavailable model gives a recoverable service error	Readiness and failure test pending
GT23	Small screen retains accessible result content	Responsive issue identified regression pending
GT24	Restored release reproduces representative paths	Backup and recovery test pending

The final report of this matrix should count executed, passed, failed, and blocked cases separately. A blocked case is not a pass. Tests should be linked to the same evaluated release unless an explicit version change explains a difference. The test record should retain failures and corrections rather than rewriting expectations after an unexpected result.

Appendix D Prospective Stakeholder Validation Protocol

This protocol is a proposed evaluation instrument and contains no participant findings. Recruitment should follow the relevant institutional and organizational requirements. Participants should be briefed on the prototype's advisory role and should not be asked to make a real diagnosis or execute an actual intervention during a software task.

Professional task 1 is to select a period and identify the unit and input origin of a prediction. Task 2 is to inspect an explanation and identify which class it explains. Task 3 is to distinguish district prediction from proxy area context. Task 4 is to review an intervention and recipient mapping in a controlled warning scenario. Task 5 is to recover from cancellation, timeout, or partial storage failure.

Inspection task 1 is to submit an authorized image and interpret the detector output. Task 2 is to identify a missed or uncertain object and explain the need for physical verification. Task 3 is to describe why no detection cannot confirm a safe premise. These tasks assess use of the tool rather than only agreement with a detector label.

Information task 1 is to ask a dengue question in a supported language form. Task 2 is to identify the cited evidence and explain any clarification or review status. Task 3 is to recognize a request for current data and assess whether the system actually has a verified current source. Expert reviewers should separately assess relevance, correctness, faithfulness, and unsafe implications.

For each task, the observer should record completion, assistance, time, error type, and the participant's explanation of the result. The facilitator should avoid leading the participant toward an intended answer. Language form, device context, and professional role should be recorded at an appropriate level without unnecessary personal data.

Post-task prompts can ask whether the result was understandable, whether its uncertainty was clear, whether the evidence was easy to inspect, and whether the workflow fit the participant's task. These are bespoke prompts. They should not be converted to a standardized usability score without using the corresponding validated instrument and scoring procedure.

The analysis should distinguish professional and public-user findings and should examine misunderstandings rather than report only average satisfaction. A small convenience sample should be described as such. No general population benefit should be inferred solely from positive opinions about a prototype.

Appendix E Academic Integrity and AI Assistance

The four project team members are the authors of this common report and are responsible for the research and component development presented within it. Generative AI was used as a supporting tool for selected images, report organization, language editing, and document formatting. The research contributions and reported experimental findings remain attributable to the team members and their respective component records.

The authors retain responsibility for accuracy, originality, references, code, data, contribution allocation, and final conclusions. Planned tests remain identified as prospective, and differing experimental records are presented separately. The final report and disclosure will be reviewed by the team members and supervisors in accordance with the applicable institutional requirements.

Appendix F Common Model and Evidence Reconciliation Record

Table F.1 identifies the research facts that require artefact-level reconciliation before the complete release can have one reproducible benchmark. It records uncertainties as substantive evidence needs rather than blank experimental values.

Table F.1 Outstanding common evidence reconciliation

Item	Current documented position	Required resolution
Prediction training period	C1 2022–2024 C2 2015–2024	Link each account to an exact dataset and checkpoint
Prediction metrics	C1 84.31% C2 86.35% accuracy	Identify final accepted experiment without averaging
Geographic unit	District model and separate area-week account	Verify target unit and area provenance
Feature inventory	Humidity lag appears in implementation list	Extract ordered schema from selected artefact
Risk thresholds	Low Medium High named	Confirm target rules and any area-state thresholds
SHAP evidence	Integration described no final vectors in common record	Retain actual attributions and consistency check
Intervention model	Preliminary state/action/reward framework	Confirm policy parameters and measured behaviour
Vision study	Inference path reported metrics unreported	Provide inventory independent split and benchmark
Stakeholder findings	No formal group results established	Execute protocol and retain analysis
Common release	Separate deployment and development contexts	Identify release and complete acceptance evidence

Resolving these items may change parts of the common evaluation. Any corrected result should identify its experiment and replace the associated interpretation consistently. A later model release should not silently alter the historical record of the experiments summarized here.